Bassez 2021 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood iLISI 1.6→7.8 of 34 samples (θ sweep 1/2/5/10, mixing rises to a plateau). One connected component, 0 detached islands. Samples intermix across one continuous strip — GO.

Gate 2 — module geography → GO (non-branching)

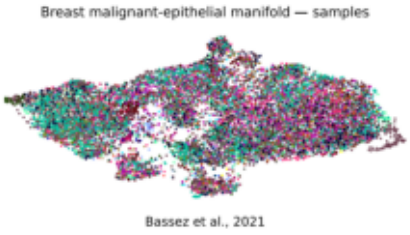
Gast & pSom cross-repress (smoothed r=−0.70; node-cluster r=−0.55) with NO co-expression hub (both-high enrichment 0.11<1). Crest opposes pSom (r=−0.91) as an inverse terminal gradient (NC-high decile pt=0.82; joint pSom&Crest 0.25<1). A single cross-repressive (Gast/pSom)→Crest axis, not two diverging arms — GO.

Gate 3 — cross-repression → NA

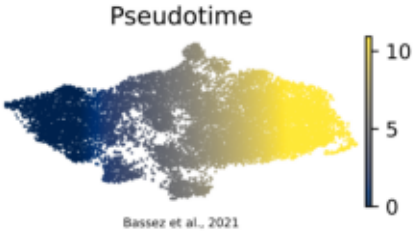
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree tips all pSom-high/Gast-low — no Gast arm). No bifurcation to test — NA.

STEP FIGURES

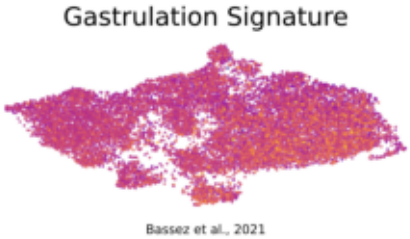
Manifold — samples



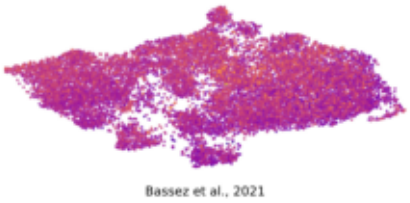
Pseudotime



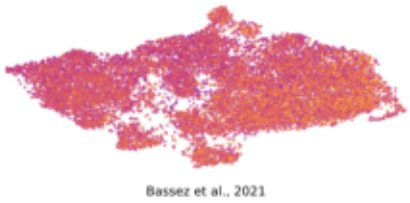
Gastrulation



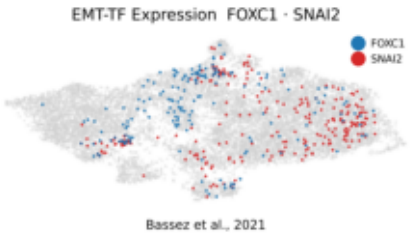
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Bassez et al., 2021), 10x — 13,834 malignant-epithelial cells (scoped from 65,885 CCCA "Malignant", 21% epithelial) across 34 samples (≥20 cells). Non-branching Linear.

Verdict.
The embryonic pattern reproduces on the correctly-scoped malignant epithelium as a single gastrulation/somitogenesis → neural-crest axis with organised Gast/pSom cross-repression (non-branching); no co-expression hub diverging into two arms, so Gate 3 is NA. Geometry Linear. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus).

Parameter choices

Scoping	CCCA "Malignant" 21% epithelial → scoped to 13,834 epithelial cells	**Threshold**	≥20 epithelial cells/sample (re-derived); 34/42 samples
Embedding	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 2,211 HVG	**Trajectory**	scFates curve (non-branching), N=20, μ=100; fold-free (2.98)
Root	pSom-high, Crest-low tip (pSom early, Crest late); ρ pSom −0.34, Crest +0.44	**Geometry**	Linear; raw means Gast −0.027 · pSom +0.030 · Crest −0.011
EMT-TFs	FOXC1 (early/root) → SNAI2 (late/terminus); VIM excluded		